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Mental Health Crisis & Socioeconomic Factors

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CERTIFICATE

This is to certify that the report titled ***Mental Health Crisis & Socioeconomic Factors*** *is a bonafide record of work done by* ***Mahima Sarah Johnson (2441439), Lahari Pallekala (2441450) & Jeffin J John (2441428)*** *of CHRIST (Deemed to be University), Bangalore, in partial fulfilment of the requirements of V Semester BSc (Data Science, Statistics) during the year 2026-27.*

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ABSTRACT

This report examines whether income inequality predicts the prevalence of mental health disorders across 28 Indian states, using the urban-rural gap in Monthly Per Capita Consumption Expenditure (MPCE) as a parameter within-state inequality. It focuses on one of the secondary objectives from a three-part team project on socioeconomic disparity and mental health burden in India titled Income Inequality and Mental Health Disorder Prevalence. Working from a state-level dataset compiled from the National Mental Health Survey (NMHS), the Household Consumption Expenditure Survey (HCES 2023-24), and related government sources, the analysis applies correlation, multiple linear regression, one-way ANOVA, and K-Means clustering. Across every test, the MPCE gap shows no statistically significant relationship with mental disorder prevalence (Pearson $r = -0.247$, $p = 0.206$; OLS $R^2 = 0.061$-0.241, $p > 0.20$ throughout; ANOVA $F = 1.38$, $p = 0.27$). Where a pattern appears, it runs opposite to the direction the global literature predicts, with more unequal states reporting marginally lower, not higher, prevalence. The report states that this null result should not be read as evidence that inequality is unrelated to mental health in India. More than half of the prevalence values in the underlying dataset are estimated by regional proxy rather than directly surveyed, which likely compresses genuine variation, and a sample of 28 states carries limited statistical power. The report concludes with a precise account of what the current data can and cannot show, and recommends that future work prioritize better-measured, longitudinal mental health data over more elaborate modelling of the same limited dataset.

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1.Introduction

Mental health disorders now sit among the leading causes of disability worldwide, cutting across age, gender, and income group. Depression, anxiety, schizophrenia, and bipolar disorder together affect more than a billion people, yet mental health remains one of the most underfunded corners of global healthcare. The COVID-19 pandemic made this shortfall harder to ignore, as health systems built for physical illness struggled to respond to a surge in psychological distress.

What drives this gap is not purely clinical. A growing body of research ties mental health outcomes to the conditions people live in: poverty, unemployment, inequality, and uneven access to care. Seen this way, mental health stops being a narrow medical concern and becomes a social and economic one, which is why it calls for the same statistical scrutiny normally reserved for economic indicators.

The scale of the problem is stark. High-income countries spend as much as USD 65 per person on mental health each year; low-income countries spend closer to four cents. That gap in resourcing does not track need, since mental illness does not spare poorer countries or poorer people within any country. If anything, the treatment gap runs along the same fault lines as income and opportunity: places with fewer psychiatrists, psychologists, and community health workers consistently show wider gaps between the number of people who need care and the number who get it.

India illustrates this tension clearly. The country ranks 135th out of 150 nations on the Gini-adjusted Human Development Index, a sign of how much inequality erodes the benefits of aggregate growth. States with high GDP per capita, such as Goa and Delhi, still report rising rates of depression and anxiety, which complicates the assumption that wealth alone insulates a population from psychological distress. Meanwhile, the poorest 40 percent of India's population has almost no meaningful access to mental health resources, and the rural-urban divide compounds the problem: agricultural communities facing chronic debt and economic precarity carry a disproportionate share of the country's mental health burden, visible most starkly in the persistence of farmer suicides.

This project sits inside a larger team investigation into how socioeconomic disparities, spanning income inequality, healthcare infrastructure, and economic structure, shape the prevalence, treatment coverage, and economic burden of mental illness. My part of that investigation looks specifically at income inequality and disorder prevalence across Indian states. The guiding question is simple to state and harder to answer cleanly: does a state's internal income inequality, measured through the gap between urban and rural consumption levels, predict how much mental illness that state reports?

Answering this matters beyond academic interest. If inequality does predict prevalence, that is an argument for targeting mental health investment toward the states and districts where the gap between urban and rural living standards is widest, rather than distributing resources evenly or by population alone. If it does not, that is also informative. It would suggest that the story is more about access and detection than about income disparity itself, and it would push the more useful question toward why certain states report the burden they do, independent of how unequal they are internally.

1.1 The Global Mental Health Crisis

Before narrowing to India, it helps to place the problem in its global context, since the same forces that shape India's numbers show up, in different proportions, almost everywhere.

Over a billion people worldwide live with some form of diagnosable mental health condition. The majority never receive care, and the reason is not simply that treatments do not exist. Effective therapies and medications for depression, anxiety, and even more severe conditions like schizophrenia have existed for decades. What is missing is the system to deliver them: trained professionals, functioning referral pathways, and public willingness to fund mental healthcare at a level that matches its share of the overall disease burden.

That underfunding is not evenly distributed. The treatment gap follows the same socioeconomic lines that shape most other health outcomes. Income level, geographic location, and how much a government chooses to invest in health all predict how wide the gap will be in a given country. Where a state's income sits low, its mental health infrastructure tends to sit lower still, and where a rural or remote population lives far from urban hospitals, distance itself becomes a form of exclusion.

The result is a measurable coverage gap between the number of people who need mental healthcare and the number who receive anything approaching adequate treatment. Countries with fewer mental health professionals per capita show a wider version of this gap, as do countries that spend less on health overall. Neither association is surprising on its own, but taken together they describe a system in which mental healthcare access is rationed less by clinical need and more by where a person happens to be born.

1.2 The Role of Socioeconomic Inequality

The treatment gap described above is not distributed randomly across countries or regions. It follows the same socioeconomic fault lines that shape most other public health outcomes: income level, geography, and the scale of government health investment. This report's specific concern, income inequality's relationship with disorder prevalence rather than treatment access, sits one step earlier in that causal chain. Before asking why people who need care do not get it, it is worth asking whether the underlying burden of illness itself tracks inequality in the first place. That is the question this report tests directly, using Indian state-level data.

2. Literature Review

The literature on mental health and socioeconomic disadvantage is large, but it clusters around a small number of recurring findings. Fourteen sources anchor this project's review, spanning global commissions, systematic reviews, causal studies, and recent OECD and WHO reporting. Read together, they build a fairly consistent case: mental illness correlates with poverty and inequality, the treatment gap is structural rather than random, and very little of this evidence has been tested with the kind of quantitative modelling that would let researchers separate correlation from something closer to a causal story.

2.1 Structural and Global Evidence

The starting point for most of this literature is the Lancet Commission on Global Mental Health and Sustainable Development. Patel and colleagues (2018) argued that mental disorders are inseparable from the broader health and economic systems they occur within, and that the treatment gap for the most severe conditions approaches 90 percent in some settings. Their call was for scaled-up, rights-respecting services rather than a narrow clinical fix, and the Commission's framing set the tone for most work that followed it.

Thornicroft and colleagues (2016) gave that framing empirical weight. Working across 21 countries, they found that only 16.5 percent of people with major depressive disorder received care that met a minimally adequate standard. The disparity between income groups was severe: roughly one in five people in high-income countries received adequate treatment, compared with about one in 27 in low- and lower-middle-income countries. That is not a small gap. It is close to a five-fold difference in the odds of getting help, determined largely by which country a person happens to live in.

The poverty side of the relationship is documented just as thoroughly. Lund and colleagues (2010), in a systematic review of community-based studies, found that 73 percent reported a positive association between poverty indicators and common mental disorders. Their language, a "vicious cycle," captures the bidirectional nature of the problem: mental illness reduces a person's capacity to work and earn, and that reduced income in turn worsens the conditions that contribute to mental illness in the first place. Ridley and colleagues (2020) pushed this further using causal-inference methods, including randomized cash transfer programmes, and found that economic shocks cause mental illness rather than merely correlating with it, and that anti-poverty interventions can measurably improve mental health outcomes.

Inequality specifically, as distinct from poverty in absolute terms, features prominently in Wilkinson and Pickett's The Spirit Level (cited in Taylor, 2009), which argued that more unequal societies report higher rates of mental illness even after accounting for average income. Related work by Pickett and colleagues (2006), and separately by Yu and Williams (1999) and Muntaner and colleagues (2004), reaches a similar conclusion from different angles: people in lower socioeconomic positions consistently face a higher risk of depression, a finding that recurs across enough independent studies that it is hard to dismiss as a fluke of any one dataset.

Resourcing gaps compound the picture. Saxena and colleagues (2007) documented how little of most national health budgets goes toward mental health, with especially sharp urban-rural disparities in the availability of specialists. The WHO's Mental Health Atlas 2024 confirms this pattern still

holds: shortages of trained professionals, insufficient psychiatric facilities, and weak community support remain the norm rather than the exception across much of the world. Kakuma and colleagues (2011) proposed that integrating mental healthcare into primary health systems and expanding community-based delivery could help close these workforce gaps without waiting for a large influx of specialist psychiatrists, a model with obvious relevance to a country like India where specialist density is low outside major cities.

At the policy level, the World Economic Forum and Harvard School of Public Health (2011) flagged that non-communicable diseases, including mental illness, remain under-prioritized relative to their economic cost. More recent OECD work sharpens this further: the OECD's 2021 benchmarking report found that 67 percent of people who said they needed mental health care also said it was unavailable to them, and the organisation's 2026 "Society at a Glance" report tied rising income inequality to a 19 percent increase in reported mental health problems across high-income country populations, one of the more direct pieces of evidence connecting inequality trends to mental health trends over time.

2.2 Synthesis

Taken as a whole, this body of work supports three claims with reasonable consistency: mental illness clusters where poverty is highest, the gap between needing and receiving care follows the same socioeconomic lines, and inequality specifically, not just low income, appears to carry its own independent association with worse mental health outcomes. What the literature does not do, almost without exception, is test these claims using the regression, clustering, or panel-based methods that would let a reader judge how strong these relationships actually are once other factors are held constant. Most of the evidence is descriptive, cross-sectional, or drawn from qualitative synthesis. That gap is what this project, and this sub-report specifically, tries to address for the case of income inequality and disorder prevalence within India.

3. Research Gaps Identified

Every strand of the literature reviewed above shares a common limitation: it establishes association without formal statistical testing, or it tests association at the individual level without extending the analysis to the state or national level where policy decisions actually get made. The table below summarizes the specific gap in each source and how this project, together with the other two sub-analyses under the shared primary objective, addresses it.

Source	Identified Gap	How This Project Fills It
Patel et al. (2018) — Lancet Commission	Discusses structural determinants qualitatively; no regression, clustering, or time-series modelling applied.	Apply OLS regression and clustering to test the same structural hypotheses on a real dataset, converting narrative claims into reproducible quantitative evidence.
Lund et al. (2010) — Poverty & Common Mental Disorders	Individual-level meta-analysis; does not distinguish absolute poverty from relative inequality (Gini-type measures); uses pre-2016 data.	Shift the unit of analysis to the state level and separately model absolute consumption expenditure versus the urban-rural expenditure gap as distinct predictors.
Wilkinson & Pickett (2009) — The Spirit Level	Bivariate correlations on a small OECD sample; no multivariate controls, no disorder disaggregation, no longitudinal component.	Apply multivariate regression with controls and extend the analysis to a non-OECD, within-country setting (India) rather than a small cross-national sample.
Thornicroft et al. (2016) — Global Treatment Gap	Describes the treatment gap without modelling its predictors; single cross-sectional wave; covers major depressive disorder only.	Test structural predictors of prevalence and, in the wider team project, treatment coverage, using regression rather than description alone.
Saxena et al. (2007) — Mental Health Resources	Descriptive census of resource scarcity; no regression testing whether resource levels predict outcomes; predates modern harmonised datasets.	Treat consumption expenditure and inequality measures as regression predictors of prevalence, using a recently compiled state-level dataset.
WHO Mental Health Atlas series (2001-2024)	A reporting instrument, not an analytical one; catalogues system characteristics without testing interactions between	Use Atlas-style indicators alongside inequality data and apply statistical modelling, including panel-style

Source	Identified Gap	How This Project Fills It
	variables or exploiting its own multi-wave structure.	comparison across waves, that the source itself never performs.
Kakuma et al. (2011) — Mental Health Workforce Gap	Cost-projection modelling only; does not empirically test whether observed workforce levels predict treatment coverage or economic outcomes.	Use observed workforce and infrastructure indicators as predictors in regression models, quantifying relationships empirically rather than projecting them theoretically.
Ridley et al. (2020) — Poverty, Depression & Anxiety	Individual-level causal inference (RCTs, quasi-experiments); cannot generalise to cross-national or cross-state structural variation.	Complement this micro-level causal evidence with a macro-level, state-as-unit structural analysis of how socioeconomic architecture relates to outcomes.
World Economic Forum & Harvard School of Public Health (2011)	A forward-looking projection exercise; does not empirically test whether current mental illness burden associates with observed economic outcomes.	Use historical, observed state-level data to test these relationships empirically rather than relying on projected assumptions.
OECD (2021, 2026)	Normative and descriptive reporting; argues for prevention investment and documents rising inequality without testing state- or country-level predictors formally.	Apply regression and group-comparison methods to test whether the inequality-outcome relationship the OECD describes narratively also holds in a formally tested, state-level Indian dataset.

Table 1. Summary of Research Gaps Identified in the Literature and How This Project Addresses Them

These six sources are representative of the fourteen reviewed in Section 2; the remaining eight share the same basic limitation, description without formal statistical testing, and are addressed by the same shift in method applied here: moving from narrative or bivariate claims to regression, group comparison, and clustering run on an actual dataset.

4. Objectives

4.1 Primary Objective

To statistically investigate how socioeconomic disparities, including income inequality, healthcare infrastructure, and economic structure, shape the differential prevalence, treatment coverage, and economic burden of mental health disorders, using open real-world datasets and rigorous statistical modelling.

This primary objective is shared across the three sub-analyses carried out by the team (Mahima, Lahari, and Jeffin), each of which isolates one link in the chain running from socioeconomic condition to mental health outcome. This report covers the first of those links.

4.2 Secondary Objective

Sub-Topic: Income Inequality & Mental Health Disorder Prevalence

To examine whether within-state income inequality, measured through the urban-rural Monthly Per Capita Consumption Expenditure (MPCE) gap, is statistically associated with the prevalence of mental health disorders across Indian states, and to identify distinct clusters of states based on their socioeconomic and mental-health profiles, in order to determine whether India's most economically unequal states also carry the highest mental health burden.

The case for this objective rests on a few observations specific to India. First, the country ranks 135th out of 150 nations on the Gini-adjusted Human Development Index, which signals that inequality here is not a marginal statistical footnote but a defining feature of the economy. Second, high-GDP states such as Goa and Delhi report rising rates of depression and anxiety despite their wealth, which challenges any simple story where prosperity alone protects mental health. Third, the poorest 40 percent of the population has close to no meaningful access to mental health services, meaning that any inequality effect is likely to interact with an access effect rather than operate independently of it. Fourth, the rural-urban divide carries its own well-documented burden, visible in the persistence of farmer suicides tied to agricultural debt and economic precarity.

Framed this way, the objective is not simply to run a correlation and report a coefficient. It is to establish, as a foundation for the team's later work on treatment access and economic burden, whether inequality itself is a meaningful predictor of prevalence in the Indian context, or whether the relationship is weaker, more complicated, or masked by how the underlying data are collected. That distinction matters for what comes next in the broader project: if inequality drives prevalence directly, policy should target unequal states; if it does not, the more useful lever may be elsewhere, in access and detection rather than income disparity itself.

Expected Outcomes

- A statistically validated conclusion, using correlation, regression, and group-comparison tests, on whether income inequality predicts mental disorder prevalence across Indian states.

- A cluster analysis grouping states by their combined inequality and mental-health-burden profile, useful for identifying which states share a common socioeconomic and mental-health signature.
- An honest account of whether the data support the poorer, more unequal states carry a higher burden hypothesis drawn from the literature, including a clear-eyed discussion of what to make of it if the data do not support that hypothesis.

5. Data Description

5.1 Data Sources

This sub-objective draws on secondary, state-level data compiled from publicly available government reports and national surveys. Four sources were merged into a single working dataset:

- The National Mental Health Survey (NMHS), conducted by NIMHANS, which remains India's most comprehensive state-level mental health prevalence survey despite dating to 2015-16.
- The National Sample Survey Office (NSSO) and Ministry of Statistics and Programme Implementation (MoSPI), specifically the Household Consumption Expenditure Survey (HCES) 2023-24, used for rural and urban MPCE figures.
- "Health of the Nation's States" reports, used to cross-check and, where necessary, proxy figures for states not directly surveyed by the NMHS.
- World Bank and Government of India socioeconomic statistics, used for broader context on regional classification and income categorisation.

5.2 Variables

The cleaned working dataset covers 28 Indian states and union territories. The table below lists each field, its meaning, and its role in the analysis.

Variable	Description	Role
State / Region	State (or UT) name and its broader geographic region (South, North, East, West, Central, Northeast).	Identifier / grouping
Rural_MPCE	Rural Monthly Per Capita Consumption Expenditure in rupees, used as a proxy for rural income level.	Predictor
Urban_MPCE	Urban Monthly Per Capita Consumption Expenditure in rupees, used as a proxy for urban income level.	Predictor
MPCE_Gap_Percent	Percentage gap between urban and rural MPCE; the primary proxy for within-state income inequality.	Primary predictor
Income_Level	Categorical grouping of states into Low, Medium, or High inequality, based on the MPCE gap.	Grouping variable
Mental_Disorder_Prevalence	Reported or estimated prevalence of mental health disorders for the state, in percent.	Outcome (DV)

Variable	Description	Role
Income_Data_Status	Flag indicating whether the MPCE figure is Actual (HCES-surveyed) or Estimated (regional proxy).	Data-quality flag
Mental_Health_Status	Flag indicating whether the prevalence figure is Actual (NMHS-surveyed) or Estimated (regional proxy).	Data-quality flag
Notes	Free-text documentation of how estimated values were derived, e.g. proxied from a neighbouring state.	Documentation

5.3 Data Quality Discussion

A closer look at the data-quality flags matters more here than in a typical dataset, because it directly shapes how the results in Section 8 should be read. Of the 28 states, 19 have an Actual, HCES-surveyed MPCE figure, while the remaining 9, mostly smaller Northeastern states plus Goa, rely on a regional proxy. On the mental health side, the split is worse: only 12 states have an Actual NMHS-surveyed prevalence figure; the other 16 are Estimated, typically as the average of nearby NMHS-surveyed states with a similar socioeconomic profile.

That is worth sitting with. Well over half of the mental health prevalence values in this dataset are not measurements at all, but regional averages standing in for a measurement that was never taken. This is not a flaw introduced by this project; it reflects a genuine limitation of India's mental health survey infrastructure, where the 2015-16 NMHS covered only 12 states directly and left the rest to be approximated by researchers and government agencies ever since. But it has a direct statistical consequence: any state whose prevalence figure is proxied from its neighbours will, almost by construction, look similar to those neighbours, which can artificially inflate regional clustering and dampen genuine state-to-state variation. This point is revisited in the Discussion and Limitations sections, because it turns out to matter a great deal for how the regression results in Section 8 should be interpreted.

6. Data Cleaning and Preparation

The raw data arrived with inconsistent column naming and mixed data types, typical of a dataset assembled by hand from several government sources. Column names were standardised to snake_case, and the four continuous fields, Rural_MPCE, Urban_MPCE, MPCE_Gap_Percent, and Mental_Disorder_Prevalence, were coerced to numeric type to catch any stray text or formatting artifacts left over from the source spreadsheets.

An initial missing-value check found up to 16 missing entries scattered across the analytic columns, including 3 missing State labels and 7 missing Region labels, concentrated among states and union territories that neither the NMHS nor the HCES surveyed directly and for which no reliable regional proxy could be constructed. Rows with an unresolved Region were dropped rather than imputed, since geography is a categorical field with no sensible numeric substitute. That left a working dataset of 28 states with zero missing values across every analytic column, which is the dataset used throughout the results reported in Section 8.

An IQR-based outlier check (the standard 1.5 times interquartile range rule) was then applied to the four continuous variables. Rural_MPCE flagged one outlier, Sikkim, at ₹9,474 per month, well above the national state-level average. Urban_MPCE flagged two, Goa at ₹10,000 and Sikkim again at ₹13,965. Neither MPCE_Gap_Percent nor Mental_Disorder_Prevalence produced any outliers under this rule. In both flagged cases, the outlier reflects a genuinely high-consumption state rather than a data-entry error: Goa and Sikkim are consistently among India's wealthiest states by per-capita measures, and their inclusion is economically meaningful rather than noise. Both were retained in the analysis. Removing them would have artificially narrowed the range of the income variables and, given the very small sample of 28 states, would have cost more in lost variation than it gained in tidiness.

7. Research Methodology

All analysis was carried out in Python, using Pandas and NumPy for data handling, SciPy and Statsmodels for statistical testing, and Scikit-learn for clustering. The workflow follows a standard sequence: describe the data, check the assumptions each test requires, test the core relationship in more than one way, and only then move to the more exploratory clustering step. The full, runnable code is included as an appendix and is written to execute in Google Colab without modification, aside from uploading the dataset.

7.1 Descriptive Statistics

Descriptive statistics were computed for all four continuous variables, namely MPCE_Gap_Percent, Rural_MPCE, Urban_MPCE, and Mental_Disorder_Prevalence. These included the mean, median, standard deviation, minimum, maximum, and skewness of each variable, and served to summarise the distribution of each variable before any formal test was applied.

7.2 Shapiro-Wilk Normality Test

A Shapiro-Wilk normality test was conducted on MPCE_Gap_Percent and Mental_Disorder_Prevalence to check whether the inequality proxy and the prevalence variable are approximately normally distributed. The outcome of this test determined whether Pearson or Spearman correlation would be more appropriate for the subsequent correlation analysis.

7.3 Correlation Analysis (Pearson and Spearman)

Correlation analysis was carried out between MPCE_Gap_Percent and Mental_Disorder_Prevalence using both the Pearson and Spearman methods, to measure the strength and direction of the association between the inequality proxy and prevalence. Pearson correlation was reported as the primary test given the approximate normality of both variables, while Spearman correlation was reported alongside as a rank-based check.

7.4 Multiple Linear Regression (OLS)

Multiple linear regression (OLS) was used to evaluate the effect of the inequality proxy on prevalence while holding rural and urban consumption levels constant, and separately while adding the Income_Level category as a control. The dependent variable in this model was Mental_Disorder_Prevalence, and the independent variables were MPCE_Gap_Percent, Rural_MPCE, Urban_MPCE, and Income_Level.

7.5 One-way ANOVA and Kruskal-Wallis

A one-way ANOVA and a Kruskal-Wallis test were applied to Mental_Disorder_Prevalence as grouped by Income_Level, to test whether mean prevalence differs significantly across the Low, Medium, and High inequality groups. The Kruskal-Wallis test was reported as a non-parametric check given the small group sizes.

7.6 Levene's Test

Levene's test was conducted on Mental_Disorder_Prevalence as grouped by Income_Level to check the equal-variance assumption that the one-way ANOVA relies on, before that result was treated as reliable.

7.7 K-Means Clustering

K-Means clustering was applied to MPCE_Gap_Percent, Rural_MPCE, Urban_MPCE, and Mental_Disorder_Prevalence, after standardising all four continuous variables, to group states with similar socioeconomic and mental-health profiles and to identify whether a distinct high-inequality, high-burden cluster of states exist

Two points about this design are worth flagging up front. First, with only 28 observations, this is a small-sample analysis by any standard, and the statistical power to detect anything but a fairly large effect is limited. Second, because roughly half of the prevalence values are estimated rather than directly measured (Section 5.3), the results below need to be read as a description of what the available data show, not as a definitive test of the underlying biological or social relationship. Both points are treated as first-class considerations throughout the results and discussion, rather than as an afterthought buried in a limitations section.

8. Results

This section reports what the data actually show, test by test, in the order laid out in the Methodology. The headline result can be stated plainly before working through the detail: across every test applied, the urban-rural MPCE gap does not show a statistically significant relationship with mental disorder prevalence in this 28-state dataset. That is not the direction the literature review in Section 3 would predict, and Section 9 spends real space on why that might be, rather than treating it as a footnote.

8.1 Descriptive Statistics

Variable	Mean	Median	Std Dev	Min	Max	Skew
Rural_MPCE (₹)	4,769.46	4,224.50	1,436.73	2,927.00	9,474.00	1.53
Urban_MPCE (₹)	7,185.75	6,956.50	1,783.29	5,114.00	13,965.00	2.16
MPCE_Gap_Percent (%)	53.61	55.45	16.94	17.40	80.60	-0.27
Mental_Disorder_Prevalence (%)	10.28	10.25	2.06	5.80	14.10	-0.16

Table 3: Descriptive Statistics Continuous Variables

The average urban-rural MPCE gap across the 28 states is 53.6 percent, meaning that in a typical state, urban households report consumption expenditure roughly one and a half times their rural counterparts. That gap ranges widely, from 17.4 percent in Kerala, where rural and urban living standards are relatively close, to 80.6 percent in Meghalaya, where the divide is far starker. Reported mental disorder prevalence averages 10.3 percent and ranges from 5.8 percent (Uttar Pradesh) to 14.1 percent (Manipur), a spread of more than eight percentage points across states.

Both Rural_MPCE and Urban_MPCE show positive skew (1.53 and 2.16 respectively), driven by the small number of high-consumption outlier states, Goa and Sikkim, discussed in Section 6. MPCE_Gap_Percent and Mental_Disorder_Prevalence are both close to symmetric, with skew values under 0.3 in either direction, which is a reasonable basis for treating them as approximately normal in the correlation step that follows.

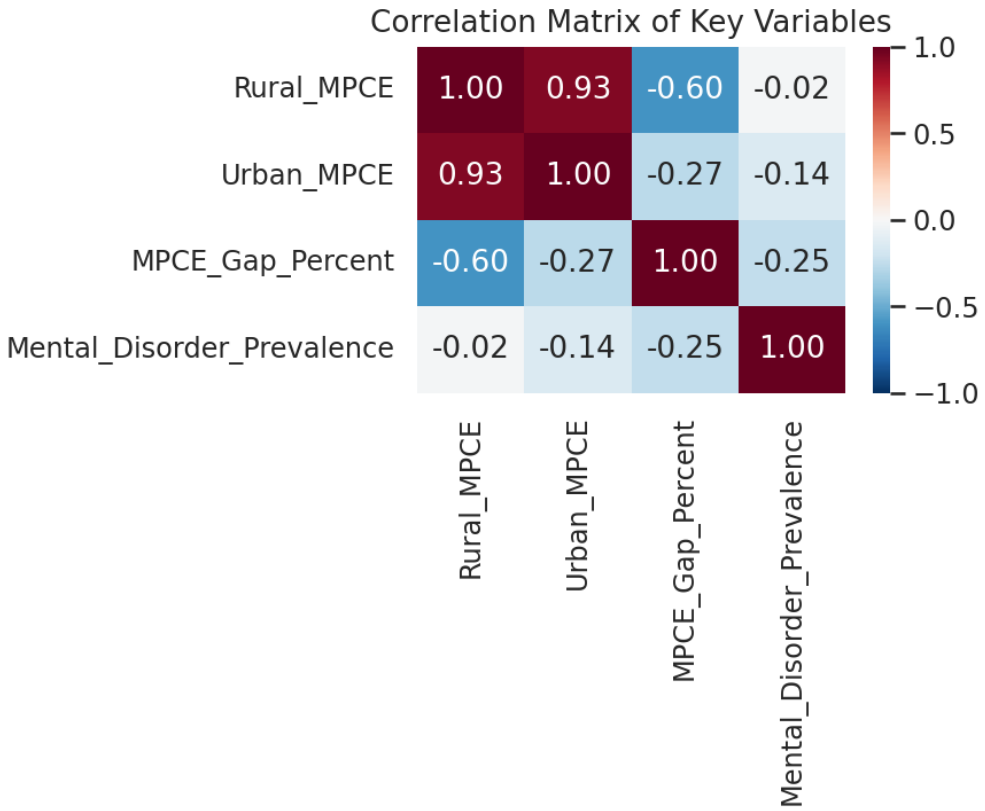


Figure 1. Correlation matrix of the four continuous variables. All relationships involving Mental_Disorder_Prevalence are weak (|r| < 0.25).

8.2 Normality Testing

Formal Shapiro-Wilk tests confirm this. Neither MPCE_Gap_Percent (W = 0.967, p = 0.495) nor Mental_Disorder_Prevalence (W = 0.970, p = 0.587) departs significantly from a normal distribution. Since both p-values sit well above the conventional 0.05 threshold, there is no statistical basis to reject normality for either variable, which supports using Pearson correlation as the primary test below, with Spearman reported alongside as a robustness check rather than a necessity.

8.3 Correlation Analysis

The Pearson correlation between the MPCE gap and mental disorder prevalence is r = -0.247 (p = 0.206). The Spearman rank correlation tells the same story: rho = -0.224 (p = 0.252). Both are weak, both point in the negative direction, meaning that as the urban-rural expenditure gap widens, reported prevalence tends to be marginally lower rather than higher, and neither comes close to conventional statistical significance.

This negative direction runs opposite to the hypothesis drawn from the literature review, where Pickett et al. (2006) and Wilkinson and Pickett (as cited in Taylor, 2009) would predict a positive relationship: more inequality, more mental illness. It is worth being precise about what this result does and does not say. A weak, non-significant negative correlation is not evidence that inequality protects against mental illness. With p-values well above 0.05 and a sample of only 28 states, the honest reading is that this dataset does not detect a reliable linear relationship between the MPCE

gap and reported prevalence in either direction. Section 9 discusses several reasons this might be, chief among them the data-quality issue flagged in Section 5.3.

For completeness, prevalence was also correlated separately against Rural_MPCE ($r = -0.020$, $p = 0.919$) and Urban_MPCE ($r = -0.143$, $p = 0.467$). Both are essentially flat. Absolute income level, whether rural or urban, shows no meaningful association with prevalence in this dataset either, which suggests the null result on the gap variable is not simply an artefact of how the gap is constructed.

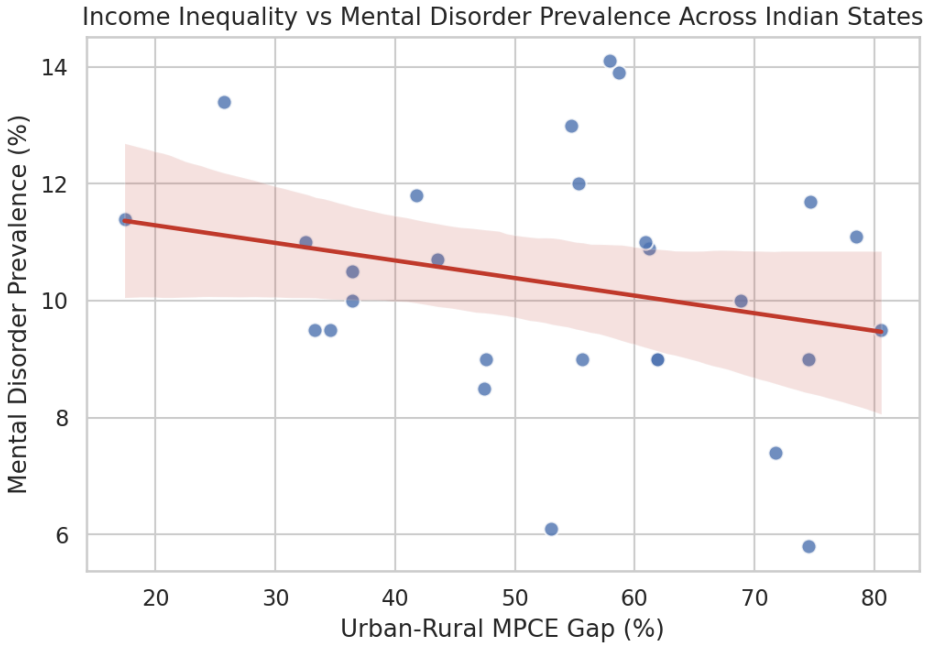


Figure 2. Mental disorder prevalence plotted against the urban-rural MPCE gap, with a fitted linear trend. The near-flat, slightly downward-sloping line reflects the weak negative correlation reported above ($r = -0.247$, n.s.).

8.4 Regression Analysis

8.4.1 Multiple Predictors (MPCE Gap, Rural MPCE, Urban MPCE)

Correlation captures a bivariate relationship; regression allows the MPCE gap to be tested while holding rural and urban consumption levels constant, which matters because those three variables are mechanically related (the gap is calculated directly from the other two). The first model regresses prevalence on MPCE_Gap_Percent, Rural_MPCE, and Urban_MPCE together.

The model explains very little of the variation in prevalence ($R^2 = 0.114$, adjusted $R^2 = 0.003$) and is not statistically significant overall ($F(3, 24) = 1.026$, $p = 0.399$). None of the three predictors reaches significance individually: the coefficient on MPCE_Gap_Percent is essentially zero ($b = -0.0007$, $p = 0.995$), Rural_MPCE is small and non-significant ($b = 0.0012$, $p = 0.721$), and Urban_MPCE is likewise non-significant ($b = -0.0010$, $p = 0.634$). The model's condition number (1.35×10^5) also flags meaningful multicollinearity between the three predictors, which is expected given that the gap is arithmetically derived from the other two, and is one reason to treat the individual coefficients here cautiously rather than as clean, independent effects.

8.4.2 Adding Income_Level and the Simple Bivariate Model

A second model adds Income_Level (the Low / Medium / High inequality categorisation) as a categorical control alongside the continuous predictors, to check whether the categorical grouping captures something the continuous MPCE gap misses. This model explains more variation ($R^2 = 0.241$, adjusted $R^2 = 0.069$) but still falls short of overall significance ($F(5, 22) = 1.398$, $p = 0.264$). The Medium inequality category comes closest to a significant difference from the High-inequality reference group ($b = 3.82$, $p = 0.074$), which is suggestive but does not clear the conventional 0.05 threshold, and with only 28 observations split across three groups and two continuous covariates, this model is almost certainly overfit relative to what the data can support.

A simpler, single-predictor model, prevalence regressed on the MPCE gap alone, confirms the same pattern in its cleanest form: $R^2 = 0.061$, $b = -0.030$ ($p = 0.205$). Less than 6 percent of the variation in state-level prevalence is explained by the urban-rural expenditure gap on its own.

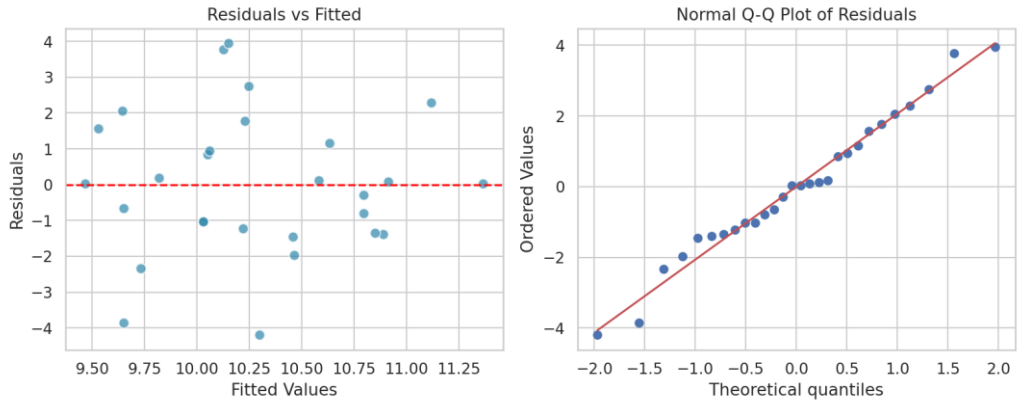


Figure 3. Residual diagnostics for the simple regression (prevalence ~ MPCE gap). The residuals show no obvious pattern against fitted values, and the Q-Q plot sits close to the reference line, so the null result is not an artefact of a poorly specified linear model.

8.5 ANOVA / Kruskal-Wallis

Grouping states into the three inequality categories already present in the dataset (Low, Medium, High) gives a second, non-parametric-friendly way to test the same relationship. Mean prevalence by group runs in the opposite direction from what the inequality hypothesis would predict: Low-inequality states average 10.87 percent prevalence (n = 9, SD = 1.24), Medium-inequality states average 10.46 percent (n = 12, SD = 2.45), and High-inequality states average the lowest of the three, 9.21 percent (n = 7, SD = 2.06).

A one-way ANOVA finds this difference is not statistically significant ($F(2, 25) = 1.381$, $p = 0.270$). A Kruskal-Wallis test, run as a non-parametric check given the modest and uneven group sizes, agrees ($H = 2.218$, $p = 0.330$). Levene's test confirms the equal-variance assumption behind the ANOVA is reasonably satisfied ($p = 0.117$), so the non-significant ANOVA result is not being distorted by unequal spread across groups. Whichever way the comparison is framed, categorical or continuous, parametric or non-parametric, the same conclusion holds: this dataset does not show a statistically reliable link between inequality category and prevalence, and what weak pattern does exist runs in the opposite direction to the literature's usual prediction.

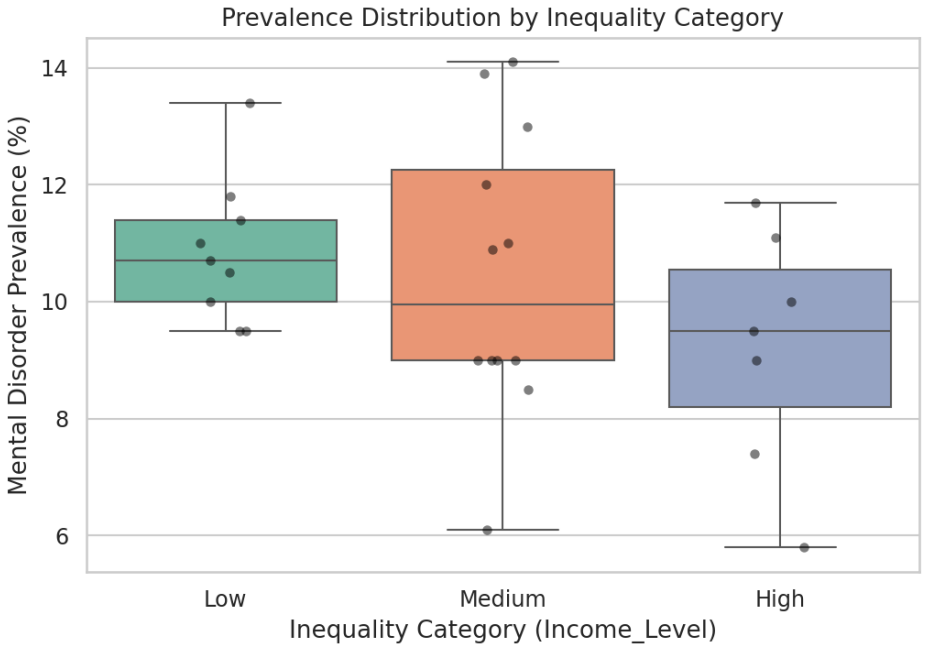


Figure 4. Mental disorder prevalence by inequality category. High-inequality states show the lowest average prevalence and Low-inequality states the highest, though the ANOVA confirms these group-differences are not statistically significant ($F = 1.38$, $p = 0.27$).

8.6 K-Means Clustering

Where the tests above ask whether one variable predicts another, K-Means clustering asks a different question: do states group naturally into distinct socioeconomic and mental-health profiles, regardless of whether any single variable drives that grouping? All four continuous variables (MPCE_Gap_Percent, Rural_MPCE, Urban_MPCE, Mental_Disorder_Prevalence) were standardised before clustering, so that variables measured in rupees do not simply dominate a variable measured in percent.

An elbow-and-silhouette analysis was run across k = 2 through k = 6 to guide the choice of cluster count. Inertia (within-cluster sum of squares) decreases steadily from 68.9 at k = 2 to 22.0 at k = 6, without a sharp elbow, which is typical for a small, continuous dataset like this one. Silhouette scores peak at k = 4 (0.353), with k = 2 (0.333) and k = 3 (0.320) close behind. Given that k = 4 splits the data into groups too small to interpret meaningfully at n = 28, and that a three-way split maps naturally onto the Low / Medium / High framing already used elsewhere in this report, k = 3 was chosen as the final solution. This is a judgment call made for interpretability rather than a case where one k value clearly dominates the others on fit alone, and it is flagged as such.

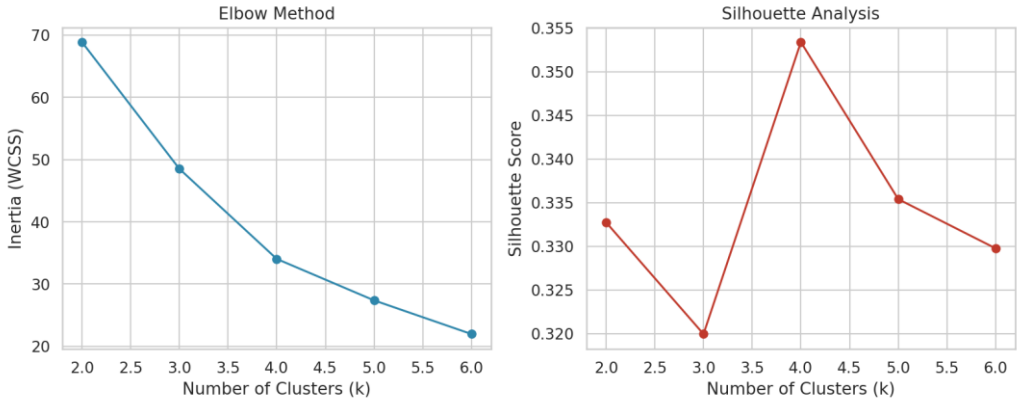


Figure 6. Elbow method (left) and silhouette scores (right) used to select the number of clusters. Silhouette scores peak at k = 4, with k = 2 and k = 3 close behind; k = 3 was selected for interpretability.

Cluster	Avg. MPCE Gap (%)	Avg. Rural MPCE (₹'000)	Avg. Prevalence (%)	Description
Cluster 0 (n = 14)	44.1	5.03	11.66	Moderate-gap, higher-prevalence — the largest group, including most of South and North India.
Cluster 1 (n = 12)	67.0	3.85	8.88	High-gap, lower-prevalence — mostly Northeast and Central/East states with wide urban-rural divides but comparatively low reported prevalence.
Cluster 2 (n = 2)	40.4	8.49	9.00	High-consumption outliers — Goa and Sikkim, distinguished mainly by very high absolute MPCE rather than by prevalence or gap.

Table 4. Description of Dataset Variables and Their Role in the Analysis

The resulting clusters reinforce, rather than overturn, the earlier finding. Cluster 1, the states with the widest urban-rural expenditure gap on average (67.0 percent), reports the lowest average prevalence of the three clusters (8.9 percent), not the highest. Cluster 0, with a more moderate gap (44.1 percent) and the lowest average rural consumption of the three clusters, reports the highest average prevalence (11.7 percent). If income inequality were a strong driver of mental illness, the expectation would be the reverse: the high-gap cluster should show the higher burden. It does not.

Cluster 2 stands apart mainly because of Goa and Sikkim's unusually high absolute consumption levels rather than because of anything distinctive about their prevalence or their gap, both of which sit close to the middle of the range across all three clusters. This is consistent with the outlier analysis in Section 6: Goa and Sikkim are unusual on income, not on mental health burden.

Cluster 1 is worth a second look given its composition: it consists largely of Northeast and Central or East India states with a notably high share of Estimated (proxied) prevalence values, as flagged in Section 5.3. Because these proxied values are often derived from averaging nearby NMHS-surveyed states, some of the apparent similarity within Cluster 1 may reflect how the data were constructed rather than a genuine shared mental-health profile. This is the clearest instance in the results where the data-quality issue raised earlier has a visible downstream effect on a specific finding, and it is one of the main reasons Section 10 treats the clustering result as suggestive rather than conclusive.

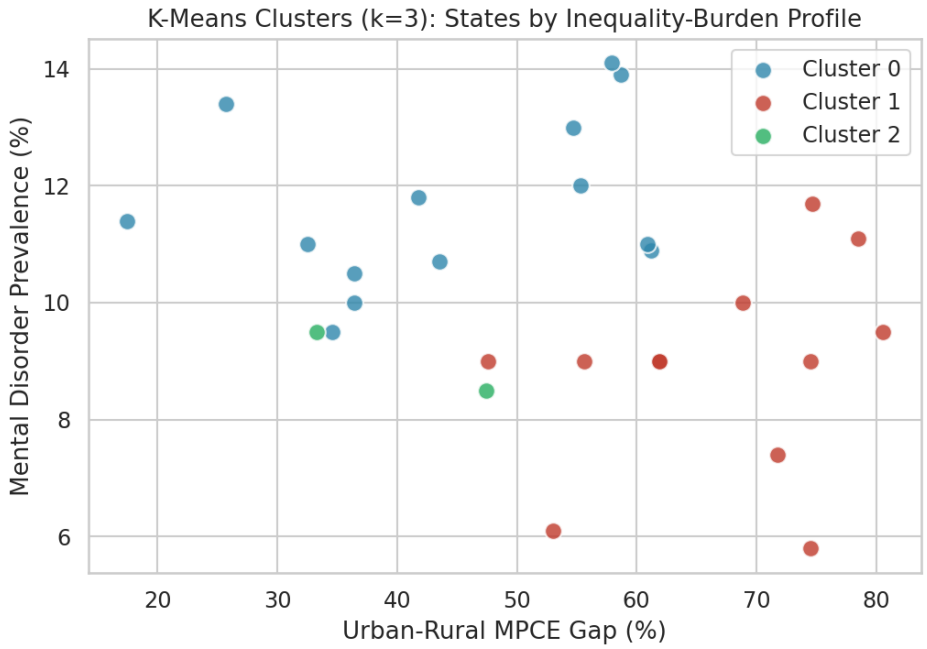


Figure 5. K-Means clusters (k = 3) plotted against the MPCE gap and prevalence. Cluster 1, with the widest average inequality gap, does not show the highest prevalence; Cluster 0 does.

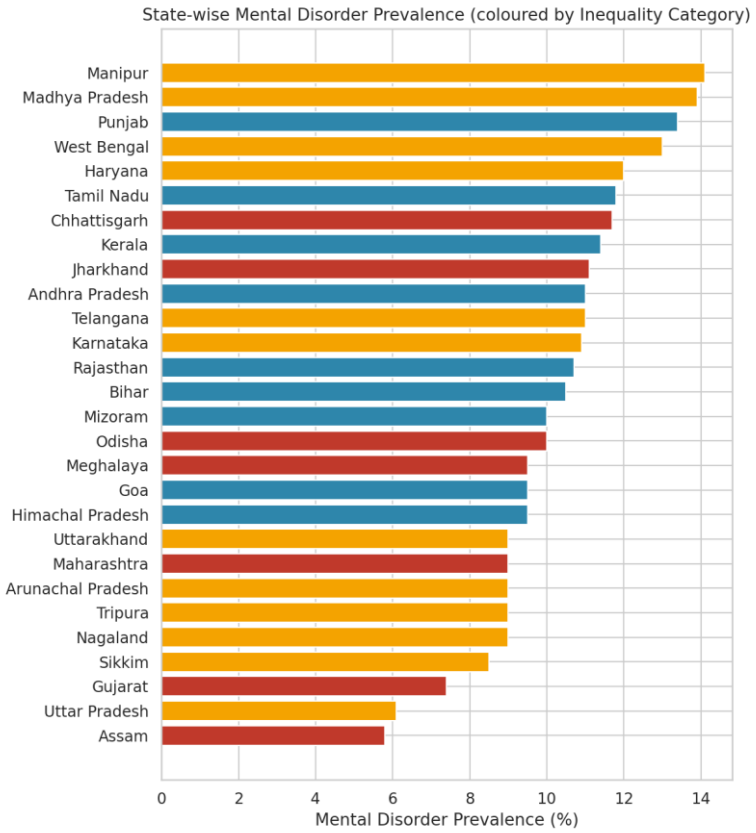


Figure 7. State-wise mental disorder prevalence, ranked and coloured by inequality category. High-inequality states (red) are not concentrated at the high-prevalence end of the chart.

8.7 Notable States

A few individual states are worth naming directly, since group averages can hide as much as they reveal. Manipur reports the highest prevalence in the dataset (14.1 percent) despite sitting in the medium inequality category with a moderate MPCE gap of 57.9 percent, and its prevalence figure is itself an Actual, NMHS-surveyed value rather than an estimate, which makes it one of the more trustworthy data points in the set. Madhya Pradesh (13.9 percent) and Punjab (13.4 percent) follow closely behind, both also Actual NMHS-surveyed states, and both fall in the Medium and Low inequality categories respectively rather than High.

At the other end, Assam reports the lowest prevalence in the dataset (5.8 percent) despite carrying the second-highest MPCE gap among all 28 states (74.5 percent) and a High inequality classification. Uttar Pradesh reports the second-lowest prevalence (6.1 percent) from a Medium-inequality classification, and Gujarat, another High-inequality state, reports the third-lowest (7.4 percent). All three are Actual, NMHS-surveyed figures, so the low reported prevalence in these states is not simply an artefact of proxying. Two of the three lowest-prevalence states in the dataset carry a High inequality classification, while all three of the highest-prevalence states carry Medium or Low classifications, which is the clearest state-level illustration of why the aggregate tests in Sections 8.3 through 8.5 came back non-significant, and in the opposite direction from the literature's usual prediction. This is not proof that inequality is irrelevant to these states' mental health outcomes; it is a reminder that a single-proxy, cross-sectional comparison can miss factors,

healthcare infrastructure, cultural reporting norms, historical survey methodology, that plausibly matter more at the level of an individual state than the inequality figure alone.

9. Discussion

The consistent theme across every test in Section 8 is a null result: no significant correlation, no significant regression coefficient, no significant group difference. Where a pattern does appear, it runs opposite to the direction the literature would predict, with higher-inequality states showing marginally lower, not higher, reported prevalence. It would be easy to read this as a straightforward finding that inequality does not matter for mental health in India. That reading would be a mistake, and this section explains why, working through each plausible explanation in turn.

9.1 Estimated Prevalence Values Likely Compress Real Variation

The most important explanation is the one flagged in Section 5.3: 16 of the 28 prevalence values, well over half, are Estimated rather than Actual, built by averaging nearby NMHS-surveyed states rather than measured directly in the state being reported on. Averaging by construction compresses variation. If a true relationship between inequality and prevalence exists but is moderate in size, a dataset where more than half the outcome values have been smoothed toward their regional neighbours is not well positioned to detect it. This is not a hypothetical concern specific to this project; it is a structural feature of working with India's mental health data, where the 2015-16 NMHS remains the only comprehensive state-level survey a decade later, and it applies to any researcher using this kind of data, not only to this analysis.

9.2 Under-Detection, Not Under-Occurrence

A related but distinct explanation is under-detection rather than under-occurrence. States with weaker mental health infrastructure, which tend to be poorer and more unequal states, often have lower survey coverage and less diagnostic capacity. A state can have a high true burden of mental illness and still report low prevalence simply because fewer people are being asked, screened, or diagnosed. If this effect is at work here, the negative correlation observed is not evidence that inequality protects against mental illness; it may be evidence that unequal states are undercounting their own burden, which would mean the true relationship, if it could be measured directly, points the opposite way from what this dataset shows.

9.3 Statistical Power at n = 28

Statistical power is a real constraint at n = 28. None of the tests in Section 8 came close to the conventional 0.05 threshold, but several sat in a range, p around 0.20 to 0.27, where a modestly larger sample or a less noisy outcome variable could plausibly tip the result toward significance. This is not a claim that a bigger sample would necessarily find what the literature predicts; it is a caution against treating "not significant at n = 28" as equivalent to "no relationship exists." The two are different claims, and this analysis can only speak to the first.

9.4 The MPCE Gap Is a Partial Proxy for Inequality

The MPCE gap is also an imperfect stand-in for the Gini-coefficient-based inequality measure most of the reviewed literature actually uses. It captures the average consumption gap between urban and rural populations within a state, which is one dimension of inequality, but it does not capture within-urban or within-rural inequality, nor does it capture wealth or asset inequality, both of which the Gini coefficient is designed to reflect more directly. A state could have a modest urban-rural gap but severe inequality within its cities, and the MPCE gap used here would not register that at all. This is a genuine limitation of the proxy, not a flaw in how it was calculated, and it means the finding here speaks specifically to the urban-rural dimension of inequality rather than to inequality in the broader, more general sense used in most of the global literature.

9.5 Cross-Sectional Design Cannot Support Causal Claims

Finally, every variable in this dataset describes a single point in time. No claim about causation, or even about a stable long-run relationship, can be drawn from a cross-sectional snapshot, no matter what the correlation coefficient says. Ridley et al. (2020), discussed in Section 3, showed that economic shocks cause mental illness using genuinely causal designs, randomized cash transfers and comparable quasi-experiments, tracked over time. A single cross-sectional comparison across 28 states cannot replicate that kind of evidence, and it was never designed to. What it can do, and what it has done here, is describe the current, static relationship between two measured quantities as they stand in the available data.

9.6 Reading the Result Honestly

Put together, these five explanations do not cancel out the null result, and they should not be read as excuses for an inconvenient finding. They are the honest boundaries of what a 28-state, single-time-point, partly-estimated dataset can tell a researcher. The most defensible conclusion is a narrow one: in this specific dataset, using this specific proxy for inequality, no statistically significant relationship with mental disorder prevalence was found, and the weak relationship that does appear runs counter to the hypothesis drawn from the global literature. Whether a genuine relationship exists that this dataset simply lacks the resolution to detect is a separate, open question that the data here cannot settle either way.

10. Limitations

- Sample size. Twenty-eight states is the full available population for this analysis, not a sample in the traditional sense, but it still limits statistical power substantially. Several results here sat close enough to conventional significance thresholds that a larger or more precisely measured dataset could plausibly shift the conclusion.
- Estimated outcome values. More than half of the prevalence figures are Estimated by regional proxy rather than directly surveyed, which likely compresses genuine variation and works against detecting a true relationship, as discussed in Section 9.2.
- Proxy validity. The urban-rural MPCE gap captures one dimension of inequality but is not equivalent to a Gini-coefficient-based measure, and does not capture within-urban, within-rural, or asset-based inequality.
- Cross-sectional design. The dataset describes one point in time, so no causal or longitudinal claim can be supported, and any apparent relationship, significant or not, should be read as descriptive rather than predictive.
- Multicollinearity. Rural_MPCE, Urban_MPCE, and MPCE_Gap_Percent are mechanically related, since the gap is calculated from the other two, which complicates interpreting individual regression coefficients in the multi-predictor models in Section 8.4.
- Cluster count selection. The choice of $k = 3$ for the K-Means analysis was made for interpretability, given that silhouette scores for $k = 2$, $k = 3$, and $k = 4$ were close together rather than showing one clearly superior solution.

11. Conclusion and Recommendations

This report set out to test whether within-state income inequality, measured through the urban-rural MPCE gap, predicts mental disorder prevalence across Indian states. Across correlation, regression, group comparison, and clustering, the answer this specific dataset gives is consistent: no. None of the relationships tested reached statistical significance, and where a weak pattern did appear, it ran opposite to what the reviewed literature would predict, with higher-inequality states showing marginally lower reported prevalence rather than higher.

This is not the finding the project set out expecting, given how consistently the global literature (Section 3) associates inequality with worse mental health outcomes. But an honest report has to follow the data rather than the hypothesis, and the most useful contribution this analysis can make is not a false confirmation of the expected story. It is a precise account of what the current India-specific, state-level data can and cannot show, and a clear diagnosis of why: a small sample, a partly estimated outcome variable, and an inequality proxy that captures only one dimension of a genuinely multidimensional concept.

For the team's broader project, this points toward two useful next steps. First, the treatment-coverage and economic-burden sub-analyses that follow this one may find a clearer signal than prevalence does here, since coverage gaps are less dependent on the same NMHS-derived prevalence estimates that likely compress variation in this analysis. Second, any future extension of this specific sub-topic would benefit most from incremental improvement in the underlying survey data itself, rather than from a more elaborate statistical model applied to the same 28 rows. A state-level Gini coefficient, a more recent and more complete mental health survey, or panel data spanning more than one time point would each, on their own, likely do more to clarify this relationship than any additional regression specification could.

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